Julian C. Stamm

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Education

Oct 2025 – present **University of Bonn**, M.Sc. in Computer Science

- Focus on Real-Time GI, Neural Rendering, and Machine Learning

Oct 2021 – Sept 2025 **University of Bonn**, B.Sc. in Computer Science

- Grade: 1.1 (top 3%)
- Bachelor's Thesis: Bidirectional Neural Radiance Caching [full-res] [low-res]

Publications

May 2026 **Photon-Driven Neural Radiance Caching**

Introduces a new photon-based training method for Neural Radiance Caching, reducing FLIP error by up to two-thirds in caustic scenes with only a ~2 ms frame-time overhead. A continuation of my Bachelor's thesis, presented at I3D 2026.
Julian C. Stamm, Tom Kneiphof, Reinhard Klein
10.1145/3807895.3807923 (I3D Companion 2026)

Experience

Feb 2026 – Mar 2026 **Teaching Assistant for "Foundations of 4D/6D Object Capture for Virtual Environments"**, University of Bonn – Bonn, Germany

- Held daily block tutorials on NeRFs, IK, 3DGS, differentiable rendering, etc.

Apr 2024 – present **Teaching Assistant for "Introduction to Computer Graphics"**, University of Bonn – Bonn, Germany

- Held weekly tutorials on rasterization, ray tracing, BSDFs, splines, etc.
- Assisted in creating exercises and exams
- Developed and maintained a modern OpenGL/C++ framework used in the exercises and the animation competition

May 2021 – Sept 2021 **Temporary Worker in Software Development Team**, Atlas Spare Parts GmbH – Ganderkesee, Germany

Developed custom CAN bus logger.

Projects

May 2026 – present **tsnn (unpublished)**

High-performance Slang-native tiny-cuda-nn alternative using Slang's auto-differentiation.

- Implemented neural spline flows, hash grid encodings, MLPs, Adam, and more
- Achieves full kernel fusion with Falcor path tracer
- Slightly outperforms tiny-cuda-nn in my NRC benchmark
- Written for my current lab research on real-time neural importance sampling

Feb 2025 – present **PhotonNRC**

Falcor implementation of the poster "Photon-Driven Neural Radiance Caching".

- Rebuilt the method as a clean Falcor render graph, improving modularity and performance
- Implemented SPPM with hardware-accelerated inverse radius search and stochastic accumulation, NRC via tiny-cuda-nn, and the SPPC training method

Feb 2025 – Sept 2025 **BiNRC**

OptiX implementation of my Bachelor's thesis "Bidirectional Neural Radiance Caching".

- Built an unbiased path tracer with MIS and NEE and a basic Disney principled BSDF
- Implemented SPPM with hardware-accelerated inverse radius search
- Implemented NRC via tiny-cuda-nn with bidirectional and photon-driven (Photon-NRC) training

Aug 2024 – present **nBounce**

Personal real-time path tracer with software BVH construction and traversal.

- Written in Rust and wgpu
- SAH-based BVH construction, Owen-scrambled RQMC sampling, VNDF importance sampling

Oct 2023 – Mar 2024 **Stable Real-Time Simulation of Mesoscopic Glints**

Computer Graphics Lab, part of my B.Sc. in Computer Science.

- Implemented the discrete stochastic microfacet model from "Real-Time Rendering of Glinty Appearances Using Distributed Binomial Laws on Anisotropic Grids" by Thomas Deliot and Laurent Belcour

Aug 2023 – present **gltemplate**

OpenGL 4.6 framework developed for CG exercises.

- Written in C++ and CMake, RAII-based resource management, texture/mesh loading utilities

Others

Apr 2026 – present **bevy_trail**

Bevy plugin for high-performance GPU-based trails in wgpu.

Sept 2025 – present **P2P Multiplayer Game**

Working on a Python-moddable P2P-multiplayer card game in Rust using bevy.

June 2023 – Sept 2023 **monarch-tree**

Submission for the animation competition in IntroCG. Procedural tree and butterfly-swarm simulation.

Skills

Languages: C++, CMake, GLSL, CUDA, Slang, Python, Rust, WGSL, Java

APIs: OpenGL, OptiX, Falcor, wgpu, bevy, Godot, PyTorch

Research Interests: Real-Time GI, Neural Rendering, Path Tracing, Photon Mapping, Machine Learning, Normalizing Flows

Spoken Languages: German (native), English (TOEFL iBT: 110/120)